Notes

Introduction to Electrostatics and Coulomb's Law

Welcome and Course Overview

Good morning everyone! We are continuing our discussion on electrostatics. For those joining for the first time, this is the second session on this chapter.

Electrostatics in NCERT

Electrostatics covers the first chapter of NCERT and approximately half of the second chapter. The concepts related to capacitors, which are also part of the second chapter, will be studied separately.

Recap of Previous Session

In the previous class, we discussed and applied **Coulomb's Law**, which describes the force acting between two charges. We will continue building upon this foundation.

Calculating Net Force on a Charge in an Equilateral Triangle

Problem Statement

We have three charges arranged at the vertices of an equilateral triangle. Let's label the vertices A, B, and C. The charges are:

- Charge at A: 2 C
- Charge at B: 4 C
- Charge at C: 4 C

The side length of the equilateral triangle is given as A. We need to find the net force acting on the 2 C charge (at vertex A).

Step 1: Identify Forces and Their Directions

We need to consider the forces exerted by the other two charges (4 C at B and 4 C at C) on the 2 C charge at A.

1. **Force due to charge at B on charge at A (let's call it F_1)**:
    - Both charges (2 C and 4 C) are positive, so the force will be repulsive.
    - Repulsive force means B will push A away from itself.
    - The direction of F_1 will be along the line joining A and B, pointing away from B (i.e., from B towards A).

- Repulsive force means C will push A away from itself.
- The direction of F_2 will be along the line joining A and C, pointing away from C (i.e., from C towards A).

Step 2: Calculate the Magnitude of Individual Forces

We use Coulomb's Law to calculate the magnitude of the forces. Coulomb's Law states that the force between two point charges is given by:

$F = k\frac{|q_1 q_2|}{r^2}$

where:

- k is Coulomb's constant
- q_1 and q_2 are the magnitudes of the charges
- r is the distance between the charges

1. **Magnitude of F_1 (between 2 C and 4 C at B)**:
   - q_1 = 2 C
   - q_2 = 4 C
   - $r = A$ (side length of the equilateral triangle)

   $F_1 = k\frac{(2\text{ C})(4\text{ C})}{A^2} = \frac{8k}{A^2}$

2. **Magnitude of F_2 (between 2 C and 4 C at C)**:
   - q_1 = 2 C
   - q_2 = 4 C
   - $r = A$ (side length of the equilateral triangle)

   $F_2 = k\frac{(2\text{ C})(4\text{ C})}{A^2} = \frac{8k}{A^2}$

Notice that $F_1 = F_2$. Let's denote this common magnitude as $F = \frac{8k}{A^2}$.

Step 3: Determine the Angle Between the Forces

Since it's an equilateral triangle, all internal angles are 60°. The forces F_1 and F_2 act along the sides of the triangle, originating from vertex A. The angle between the directions of F_1 and F_2 is the external angle at vertex A, which is $180^\circ - 60^\circ = 120^\circ$. However, the forces are directed outwards from A. The angle between the vectors F_1 and F_2 is the angle of the equilateral triangle, which is 60°.

parallelogram law for vector addition. The resultant force F_{net} of two vectors of equal magnitude F with an angle θ between them is given by:

$$F_{net} = 2F \cos\left(\frac{\theta}{2}\right)$$

In this case:

- $F = \frac{8k}{A^2}$
- $\theta = 60^\circ$ (the angle between the two force vectors)

Substitute the values:

$$F_{net} = 2\left(\frac{8k}{A^2}\right) \cos\left(\frac{60^\circ}{2}\right) \quad F_{net} = 2\left(\frac{8k}{A^2}\right) \cos(30^\circ)$$

We know that $\cos(30^\circ) = \frac{\sqrt{3}}{2}$.

$$F_{net} = 2\left(\frac{8k}{A^2}\right)\left(\frac{\sqrt{3}}{2}\right) \quad F_{net} = \frac{8\sqrt{3}k}{A^2}$$

Alternative Method: Vector Resolution

Another way to find the resultant force is by resolving the vectors into components.

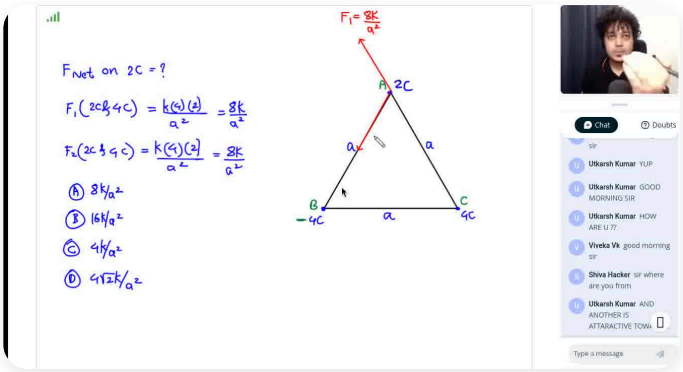
1. **Set up a coordinate system**: Let's place the 2 C charge at the origin $(0, 0)$. Let the x-axis be horizontal.
2. **Resolve F_1**: F_1 acts along the line AB. If we assume the base of the triangle is horizontal, then F_1 would be at an angle of 30° with the vertical axis (or 60° with the horizontal axis). Let's align the resultant force along the y-axis for simplicity. This means F_1 makes an angle of 30° with the negative y-axis, and F_2 makes an angle of 30° with the negative y-axis.
   - $F_{1x} = F \sin(30^\circ)$ (pointing right)
   - $F_{1y} = -F \cos(30^\circ)$ (pointing down)
3. **Resolve F_2**: F_2 acts along the line AC.
   - $F_{2x} = -F \sin(30^\circ)$ (pointing left)
   - $F_{2y} = -F \cos(30^\circ)$ (pointing down)
4. **Sum the components**:
   - **Sum of x-components**: $F_{net,x} = F_{1x} + F_{2x} = F \sin(30^\circ) - F \sin(30^\circ) = 0$
   - **Sum of y-components**: $F_{net,y} = F_{1y} + F_{2y} = -F \cos(30^\circ) - F \cos(30^\circ) = -2F \cos(30^\circ)$

Both methods yield the same result.

Final Answer

The net force acting on the 2 C charge is $\frac{8\sqrt{3}k}{A^2}$.

Quiz: Net Force with a Negative Charge



Problem Statement

Consider the same setup as the previous problem, but with a crucial change: one of the charges is now negative. Specifically, let's assume the charge at point B is now $-4C$ instead of $+4C$. The charge at point A is still $+2C$, and the charge at point C is still $+4C$. All charges are placed at the vertices of an equilateral triangle with side length a. Calculate the net force on the charge at point A.

Given:

- Charge at A: $q_A = +2C$
- Charge at B: $q_B = -4C$
- Charge at C: $q_C = +4C$
- Side length of equilateral triangle: a

Options:

A) $8k/a^2$ B) $16k/a^2$ C) $4k/a^2$ D) $4\sqrt{2}k/a^2$

Analysis of Forces

1. **Force between A and C (F_{AC}):**
    - Charges $q_A = +2C$ and $q_C = +4C$ are both positive, so the force is repulsive.
    - The magnitude of this force remains the same as in the previous scenario: $F_{AC} = \frac{k|q_A q_C|}{a^2} = \frac{k(2)(4)}{a^2} = \frac{8k}{a^2}$

- Charges $q_A = +2C$ and $q_B = -4C$ are now unlike charges, so the force is attractive.
- The magnitude of this force is: $F_{AB} = \frac{k|q_A q_B|}{a^2} = \frac{k(2)(4)}{a^2} = \frac{8k}{a^2}$
- Direction: Towards B, along the line AB.

Resultant Force Calculation

- We have two forces of equal magnitude, $F_{AC} = F_{AB} = F = \frac{8k}{a^2}$.
- The angle between these two force vectors needs to be determined.
- Since it's an equilateral triangle, the internal angles are 60°.
- The force F_{AC} acts away from C, and F_{AB} acts towards B.
- If we extend the line AC, the angle between the direction of F_{AC} and the line AB is 60°. The angle between the direction of F_{AB} (towards B) and the line AC (away from C) is $180^\circ - 60^\circ = 120^\circ$.
- Alternatively, consider the angle between the two force vectors. The force F_{AC} points along the line from A to C (repulsive). The force F_{AB} points along the line from A to B (attractive). The angle between these two lines is 60°. However, the vectors are oriented such that the angle between them is 120°. This can be visualized by placing the tails of both vectors at point A. One vector points along the side AC, and the other points along the side AB. The angle between these two vectors is 120°.

Shortcut for Resultant of Two Equal Vectors at 120°

When two vectors of equal magnitude F are inclined at an angle of 120°, their resultant F_{net} is also equal to F.

Using the formula for the resultant of two vectors: $F_{net} = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos\theta}$ Here, $F_1 = F_2 = F = \frac{8k}{a^2}$ and $\theta = 120^\circ$. $F_{net} = \sqrt{F^2 + F^2 + 2F \cdot F\cos(120^\circ)}$ $F_{net} = \sqrt{2F^2 + 2F^2(-1/2)}$ $F_{net} = \sqrt{2F^2 - F^2}$ $F_{net} = \sqrt{F^2}$ $F_{net} = F$

Therefore, the net force on charge A is $F_{net} = \frac{8k}{a^2}$.

Alternative Shortcut (using $2F\cos(\theta/2)$)

If the two vectors are equal in magnitude and the angle between them is θ, the resultant is given by $2F\cos(\theta/2)$. In this case, $F = \frac{8k}{a^2}$ and $\theta = 120^\circ$. $F_{net} = 2F\cos(120^\circ/2)$ $F_{net} = 2F\cos(60^\circ)$ $F_{net} = 2F(1/2)$ $F_{net} = F$

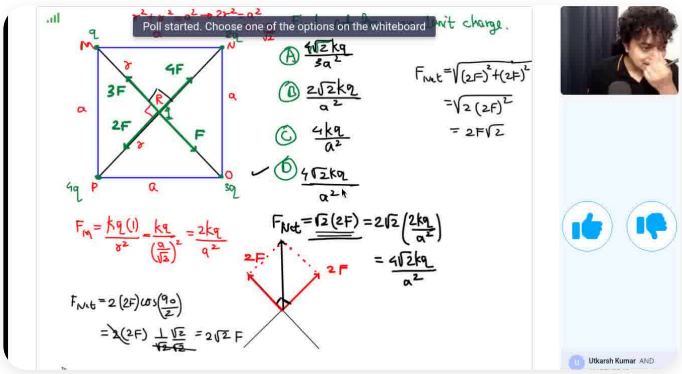
So, the net force is $\frac{8k}{a^2}$.

Conclusion

Notes

The correct option is A.

Problem: Net Force on a Unit Charge at the Center of a Square



Problem Statement

We have a square with side length 'A'. Charges q, 2q, 3q, and 4q are placed at the four corners of the square. A unit charge (1 Coulomb) is placed at the center of the square (intersection of the diagonals). The objective is to determine the net force acting on this unit charge.

Given Information

- Square side length: A
- Charges at corners: q, 2q, 3q, 4q
- Charge at center: 1 unit (1 Coulomb)

Options

A. $\frac{4\sqrt{2}Kq}{a^2}$ B. $\frac{2\sqrt{2}Kq}{a^2}$ C. $\frac{4Kq}{a^2}$ D. $\frac{4\sqrt{2}Kq}{a^2}$

Calculating the Distance from Corners to Center

Let's denote the corners of the square as M, N, O, P, and the center as R. The distance from each corner to the center of the square is the same. Let this distance be 'r'.

Consider a right-angled triangle formed by two adjacent corners and the center. The diagonal of the square has a length of $A\sqrt{2}$. The distance 'r' from a corner to the center is half of the diagonal.

Alternatively, using the Pythagorean theorem for a right-angled triangle formed by the center, a corner, and the midpoint of a side: If the side length is A, then the distance from the center to the midpoint of a side is $A/2$. The distance from the center to a corner 'r' can be found using the Pythagorean theorem on a right triangle formed by half a diagonal and two sides of length $A/2$.

$r = \frac{A\sqrt{2}}{2} = \frac{A}{\sqrt{2}}$

So, the square of the distance is: $r^2 = \left(\frac{A}{\sqrt{2}}\right)^2 = \frac{A^2}{2}$

Calculating Individual Forces

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Calculating Net Force Along Diagonals

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Calculating the Final Net Force

We now have two resultant forces, each of magnitude $2F$, acting along the two diagonals of the square. Since the diagonals of a square are perpendicular to each other, the angle between these two resultant forces is 90°.

We can find the net force using the parallelogram law of vector addition for two perpendicular vectors: $F_{Net} = \sqrt{P^2 + Q^2 + 2PQ\cos\theta}$ Here, $P = 2F$, $Q = 2F$, and $\theta = 90^\circ$. Since $\cos(90^\circ) = 0$, the formula simplifies to: $F_{Net} = \sqrt{(2F)^2 + (2F)^2}$ $F_{Net} = \sqrt{4F^2 + 4F^2}$ $F_{Net} = \sqrt{8F^2}$ $F_{Net} = F\sqrt{8} = F \cdot 2\sqrt{2}$ $F_{Net} = 2\sqrt{2}F$

Alternatively, for two equal vectors (magnitude X) at 90°, the resultant is $X\sqrt{2}$. Here, $X = 2F$, so the resultant is $2F\sqrt{2}$.

Now, substitute the value of $F = \frac{2kq}{A^2}$: $F_{Net} = 2\sqrt{2}\left(\frac{2kq}{A^2}\right)$ $F_{Net} = \frac{4\sqrt{2}kq}{A^2}$

This matches option D.

axes), the two resultant forces $2F$' would be along the axes. For example, one $2F$' along the positive x-axis and the other $2F$ along the positive y-axis (depending on the orientation of the charges).

If the two $2F$ forces are perpendicular, the resultant magnitude is simply $\sqrt{(2F)^2 + (2F)^2} = 2F\sqrt{2}$.

Alternative Method using $2A\cos(\theta/2)$ for equal vectors

When two vectors of equal magnitude 'A' are at an angle θ, their resultant is $2A\cos(\theta/2)$. Here, $A = 2F$ and $\theta = 90^\circ$. $F_{Net} = 2(2F)\cos\left(\frac{90^\circ}{2}\right)$ $F_{Net} = 4F\cos(45^\circ)$ $F_{Net} = 4F\left(\frac{1}{\sqrt{2}}\right)$ $F_{Net} = \frac{4F}{\sqrt{2}} = \frac{4F\sqrt{2}}{2} = 2\sqrt{2}F$ Substituting $F = \frac{2kq}{A^2}$: $F_{Net} = 2\sqrt{2}\left(\frac{2kq}{A^2}\right) = \frac{4\sqrt{2}kq}{A^2}$

All methods yield the same result.

Final Answer

The net force on the unit charge at the center is $\frac{4\sqrt{2}kq}{A^2}$.

Correct Answer: Option D

Introduction to Electrostatics Session 2

Session Overview

This is the second session on Electrostatics. The first session covered Coulomb's law, and this session will focus on solving new problems related to it. If you are new to the class, it is recommended to review the previous session's content to avoid difficulties. Replays of all sessions are available and can be accessed with the help of academic mentors.

Problem Setup: Simple Harmonic Motion of a Charge

We are considering a scenario with two charges, +Q and +Q, placed on the x-axis. The total distance between these two charges is $2A$, meaning each charge is at a distance A from the

The Question

The core question is whether this displaced charge $-q$ will perform Simple Harmonic Motion (SHM) or not. This involves analyzing the force acting on the charge $-q$ due to the other two charges.

Condition for Simple Harmonic Motion (SHM)

For a motion to be classified as Simple Harmonic Motion, two conditions must be met:

1. **Force and Displacement Relationship:** The restoring force acting on the object must be directly proportional to its displacement from the equilibrium position.
2. **Direction:** The restoring force must be directed opposite to the displacement.

Mathematically, this can be expressed as:

- $F \propto -y$ (Force is proportional to negative of displacement)
- $a \propto -y$ (Acceleration is proportional to negative of displacement)

Example: Simple Pendulum

Consider a simple pendulum. If you displace it to the right, the restoring force acts to the left. If you displace it to the left, the restoring force acts to the right. This demonstrates the force being opposite to the displacement.

Goal of the Problem

We need to determine if the force acting on the charge $-q$ satisfies the conditions for SHM. The problem might also be framed as identifying the correct relationship between force (F) and displacement (y) from options like:

- $F \propto Y$
- $F \propto -Y$
- $F \propto 1/Y$
- $F \propto -1/Y$

Calculating Force on a Displaced Charge

Introduction

When a charge $-q$ is displaced from the center, it experiences forces from other charges. Let's consider a scenario where $-q$ is displaced along the y-axis, and there are two positive charges, say

Notes

Force Analysis

1. **Forces are Attractive:** The force applied by the left-hand side charge ($+Q$) on $-q$ will be attractive. Similarly, the force applied by the right-hand side charge ($+Q$) on $-q$ will also be attractive.
2. **Equal Magnitudes:** Due to symmetry, the magnitudes of these two forces, let's call them F_1 and F_2, will be equal.

Calculating Force Magnitude

To calculate the value of these forces, we use Coulomb's Law: $F = \frac{k|q_1 q_2|}{R^2}$ Where:

- k is Coulomb's constant.
- q_1 and q_2 are the magnitudes of the charges.
- R is the distance between the two charges.

In our case, $q_1 = Q$ and $q_2 = q$. The distance R between $+Q$ and $-q$ can be found using the Pythagorean theorem. If the charge $+Q$ is at $(a, 0)$ and $-q$ is at $(0, y)$, then R is the hypotenuse of a right-angled triangle with sides a and y. $R^2 = y^2 + a^2$ So, the magnitude of each force (F_1 or F_2) is: $F_1 = F_2 = \frac{kQq}{y^2+a^2}$

Resultant Force Calculation

To find the net force, we need to add these two forces vectorially. Since the system is symmetric, we can use a shortcut for adding two equal forces. If the angle between F_1 and F_2 is 2θ, the resultant force F_{net} is given by: $F_{net} = 2F_1 \cos(\theta)$

From the diagram, consider the right-angled triangle formed by the origin, the position of $+Q$, and the position of $-q$. The angle θ is the angle between the line connecting $+Q$ and $-q$ and the y-axis. The adjacent side to θ is y, and the hypotenuse is $R = \sqrt{y^2 + a^2}$. Therefore, $\cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{y}{R} = \frac{y}{\sqrt{y^2+a^2}}$.

Substitute F_1 and $\cos(\theta)$ into the net force formula: $F_{net} = 2\left(\frac{kQq}{y^2+a^2}\right)\left(\frac{y}{\sqrt{y^2+a^2}}\right)$ $F_{net} = \frac{2kQqy}{(y^2+a^2)^1(y^2+a^2)^{1/2}}$ Using the laws of indices ($x^m \cdot x^n = x^{m+n}$): $F_{net} = \frac{2kQqy}{(y^2+a^2)^{3/2}}$ This is the magnitude of the net force on the charge $-q$.

Approximation for Small Displacements

If the displacement y is very small compared to a (i.e., $y \ll a$), we can make an approximation: When $y \ll a$, then $y^2 \ll a^2$. Therefore, $y^2 + a^2 \approx a^2$.

Notes

Direction of Net Force

The two attractive forces F_1 and F_2 are directed towards the positive charges. Due to symmetry, their horizontal components cancel out, and their vertical components add up. Since the charge $-q$ is displaced upwards (along the positive y-axis), the resultant force will be directed downwards (along the negative y-axis), trying to restore the charge to its equilibrium position. If we were to write this in vector form, there would be a minus sign indicating the downward direction.

Understanding Force Direction and Equilibrium

Resultant Force and Direction

When dealing with two forces of equal magnitude, the resultant force's direction is crucial. The parallelogram law can be used to determine the direction of the net force. If one force (F_1) is directed in one way and another force (F_2) in another, their resultant will be determined by the vector sum.

Displacement vs. Net Force Direction

In the discussed scenario, the displacement was upward (along the positive y-axis), but the net force was directed downward (along the negative y-axis). This opposition in direction is indicated by a negative sign in the force relation, meaning the force is proportional to $-y$.

When Force is Proportional to Positive Y

For the force to be proportional to $+y$ (i.e., the resultant force is upward, in the same direction as displacement), a change in the charge configuration is required. If the central charge were changed from $+q$ to $-q$, the magnitudes of the forces would remain the same, but their directions would alter. In this modified scenario, the resultant force would be upward, aligning with the upward displacement. This would lead to a relationship where the force is proportional to $+y$.

Homework Questions: Horizontal Displacement of Charge

Problem Description

We are given two homework questions involving a small charge 'q' placed at the center of a system. The setup is similar to previous problems, with a characteristic length 'a'. This time, the charge is displaced horizontally by a small distance 'x', where $x \ll a$.

Question 1: Positive Charge

If the charge 'q' is positive and is displaced horizontally by a small distance 'x', determine the relationship between the restoring force (or some other relevant quantity, implied as 'wf' in the transcript, likely work done or force) and 'x'. Is 'wf' proportional to 'x', x^2, or $-x$?

horizontally by a small distance 'x', what is the relationship between 'wf' and 'x'?

Instructions

Solve these two problems for homework. We will discuss the answers in the next class. The key difference from previous problems is the horizontal displacement instead of vertical displacement, and considering both positive and negative charges.

Equilibrium of Charges: Introduction

Understanding Equilibrium

Equilibrium is a fundamental concept in physics, particularly in the context of Newton's laws of motion. When a system is in equilibrium, it means that the net force acting on it is zero. If we consider a system of charges, for the entire system to be in equilibrium, every individual charge within that system must also be in equilibrium. This implies that the net force on each charge must be zero.

Conditions for Equilibrium

For a charge to be in equilibrium, the forces acting on it must satisfy two conditions:

1. **Equal Magnitude:** The forces must have equal magnitudes.
2. **Opposite Direction:** The forces must act in opposite directions.

If two forces act in the same direction, their resultant will be an addition of their magnitudes, and thus cannot be zero. Therefore, for equilibrium, it is crucial that forces are in opposing directions.

Analyzing Charge Placement for System Equilibrium

Consider a scenario with two fixed charges, $-9q$ and $+q$. The problem asks where a third charge, Q, should be placed so that the *entire system* (all three charges) is in equilibrium.

Case 1: Placing a Positive Charge ($+Q$) Between $-9q$ and $+q$

Let's analyze if placing a positive charge ($+Q$) between the two existing charges, $-9q$ and $+q$, can lead to equilibrium for the system.

Equilibrium of the Middle Charge ($+Q$)

1. **Force from $-9q$:** The charge $-9q$ is negative, and $+Q$ is positive. Opposite charges attract, so $-9q$ will exert an attractive force on $+Q$ towards the left.
2. **Force from $+q$:** The charge $+q$ is positive, and $+Q$ is positive. Like charges repel, so $+q$ will exert a repulsive force on $+Q$ towards the left.

arrangement.

Equilibrium of Charges: Exploring Possibilities

Conditions for Equilibrium

When considering the equilibrium of a charge, the net force acting on it must be zero. This implies that all forces acting on the charge must balance each other out, both in magnitude and direction.

Scenario 1: Positive Test Charge Between Two Positive Charges

Let's consider a scenario where a positive test charge ($+Q$) is placed between two other positive charges, say $9Q$ and another $+Q$. If we analyze the forces on the test charge:

- The force exerted by $9Q$ on $+Q$ would be repulsive, pushing $+Q$ away from $9Q$.
- The force exerted by the other $+Q$ on $+Q$ would also be repulsive, pushing $+Q$ away from that $+Q$.

In this configuration, both forces would act in the same direction, meaning the net force would not be zero. Therefore, equilibrium is not possible for the test charge in this arrangement.

Scenario 2: Positive Test Charge Outside Two Positive Charges

Now, let's consider placing a positive test charge ($+Q$) to the right of two positive charges, $9Q$ and $+Q$. If we analyze the forces on the test charge:

- The force exerted by $9Q$ on $+Q$ would be repulsive, pushing $+Q$ to the right.
- The force exerted by the other $+Q$ on $+Q$ would also be repulsive, pushing $+Q$ to the right.

Again, both forces act in the same direction, so equilibrium is not possible.

Scenario 3: Negative Test Charge Between Two Positive Charges

What if the test charge is negative, say $-Q$, and it is placed between two positive charges, $9Q$ and $+Q$? Let's analyze the forces on $-Q$:

- The force exerted by $9Q$ on $-Q$ would be attractive, pulling $-Q$ towards $9Q$ (to the left).
- The force exerted by $+Q$ on $-Q$ would also be attractive, pulling $-Q$ towards $+Q$ (to the right).

In this case, the forces are in opposite directions, making equilibrium possible. The exact position for equilibrium would depend on the magnitudes of the charges and the distances involved.

Scenario 4: Negative Test Charge Outside Two Positive Charges

- The force exerted by $9Q$ on $-Q$ would be attractive, pulling $-Q$ towards $9Q$ (to the left).
- The force exerted by $+Q$ on $-Q$ would also be attractive, pulling $-Q$ towards $+Q$ (to the left).

Both forces act in the same direction, so equilibrium is not possible for the test charge in this arrangement.

Conclusion on Equilibrium Possibilities

For a charge to be in equilibrium, the forces acting on it must be in opposite directions. This often requires a specific arrangement of charges, such as a negative charge placed between two positive charges, or a charge placed in a region where the forces from other charges can cancel each other out due to their directions.

Conditions for Zero Resultant Force

Placement of Third Charge

- For the resultant force on a charge to be zero, the forces acting on it must be in opposite directions.
- If a third charge is placed *off the line* joining the other two charges, the forces exerted by the original two charges will always have an angle between them (acute or obtuse).
- In such a scenario, the forces will never be exactly opposite, and thus, their resultant cannot be zero.
- Therefore, for the resultant force to be zero, the third charge *must* be placed on the line joining the other two charges.

Solving for Unknowns: Position and Value of Third Charge

Problem Statement

We need to find the position (X) and value (Q) of a third charge such that the system is in equilibrium. This requires solving two equations for two unknowns.

Equilibrium of the Charge at -q

Let's consider the equilibrium of the charge at position -q. It experiences two forces:

1. Force due to the small charge q.
2. Force due to the charge -9q.

For equilibrium, these two forces must be equal in magnitude and opposite in direction.

- Force due to small q on -q: $F_1 = \frac{k \cdot q \cdot (-q)}{X^2}$
- Force due to -9q on -q: $F_2 = \frac{k \cdot (-9q) \cdot (-q)}{(L+X)^2}$

Equating the magnitudes of the forces: $\frac{k \cdot q \cdot q}{X^2} = \frac{k \cdot 9q \cdot q}{(L+X)^2}$

Cancel out common terms ($k \cdot q \cdot q$): $\frac{1}{X^2} = \frac{9}{(L+X)^2}$

Take the square root on both sides: $\frac{1}{X} = \frac{3}{L+X}$

Cross-multiply: $L + X = 3X \; L = 3X - X \; L = 2X \; X = \frac{L}{2}$

So, the position of the charge -q is at a distance of $L/2$ from the small charge q.

Determining the Value of the Third Charge (Q)

Equilibrium of the Small Charge q

Now, let's consider the equilibrium of the small charge q. It experiences forces from the charge -9q and the capital charge Q.

- Force due to -9q on q: $F_3 = \frac{k \cdot (-9q) \cdot q}{L^2}$
- Force due to Q on q: $F_4 = \frac{k \cdot Q \cdot q}{X^2}$

Equating the magnitudes of the forces: $\frac{k \cdot 9q \cdot q}{L^2} = \frac{k \cdot Q \cdot q}{X^2}$

Cancel out common terms ($k \cdot q$): $\frac{9q}{L^2} = \frac{Q}{X^2}$

Substitute the value of $X = L/2$: $\frac{9q}{L^2} = \frac{Q}{(L/2)^2} \; \frac{9q}{L^2} = \frac{Q}{L^2/4} \; \frac{9q}{L^2} = \frac{4Q}{L^2}$

Cancel out L^2 from both sides: $9q = 4Q \; Q = \frac{9q}{4}$

Polarity of the Charge Q

We assumed Q to be positive during the calculation of its magnitude. Let's analyze the forces on the small charge q if Q is positive:

- Force from -9q on q: Attractive (towards -9q, i.e., to the right).
- Force from +Q on q: Repulsive (away from +Q, i.e., to the right).

If both forces are to the right, the resultant force cannot be zero. Therefore, for equilibrium, the charge Q must be negative.

Final Answer

The value of the third charge Q must be negative, and its magnitude is $9q/4$. $Q = -\frac{9q}{4}$

exercise: place the charge on the left side of the system and analyze the forces to determine its position and value.

Coulomb's Law in Vector Form

Introduction to Coulomb's Law in Vector Form

Coulomb's Law describes the electrostatic force between two charged particles. When dealing with forces, it is often necessary to consider their directional properties, which is where the vector form of Coulomb's Law becomes essential.

Consider two charges, Q_1 and Q_2, separated by a distance R. To express the force between them in vector form, we need to account for both the magnitude and the direction of the force. This concept is best understood through examples, which will be explored further.

Calculating Electrostatic Force in Vector Form

Problem Statement

We are given two points, A and B, with charges and coordinates:

- **Point A:** Charge q_1 = −5 Coulomb
- **Point B:** Charge q_2 = +5 Coulomb

We need to find the force acting on the charge at point B (+5 C) due to the charge at point A (−5 C).

Identifying Victim and Culprit

In such problems, it's crucial to identify the 'victim' (the charge on which the force is acting) and the 'culprit' (the charge causing the force).

- **Victim:** +5 C (at point B)
- **Culprit:** −5 C (at point A)

Defining the Position Vector (r-bar)

The position vector $\vec{r}$ (or $\vec{r}_{AB}$) is drawn from the culprit to the victim. It starts from the charge applying the force and ends at the charge on which the force is applied.

In this case, $\vec{r}$ starts from A and goes to B. Therefore, $\vec{r} = \vec{AB}$.

To calculate $\vec{AB}$, we use the formula: $\vec{AB} = \vec{B} - \vec{A}$ (final position vector minus initial position vector).

- Point B: $(7, 5, 8)$

So, $\vec{A} = 4\hat{i} + 5\hat{j} + 4\hat{k}$ And $\vec{B} = 7\hat{i} + 5\hat{j} + 8\hat{k}$

Now, calculate $\vec{r}$: $\vec{r} = (7 - 4)\hat{i} + (5 - 5)\hat{j} + (8 - 4)\hat{k}$ $\vec{r} = 3\hat{i} + 0\hat{j} + 4\hat{k}$ $\vec{r} = 3\hat{i} + 4\hat{k}$

Calculating the Magnitude of r-bar

The magnitude of $\vec{r}$, denoted as r, represents the distance between the two charges. $r = |\vec{r}| = \sqrt{(3)^2 + (4)^2}$ $r = \sqrt{9 + 16} = \sqrt{25}$ $r = 5$ units

Coulomb's Law in Vector Form

When calculating the electrostatic force in vector form, we use a modified version of Coulomb's Law. This form inherently includes both magnitude and direction, and it requires considering the signs of the charges.

The vector form of Coulomb's Law is: $\vec{F} = k\frac{q_1 q_2}{r^3}\vec{r}$

Key Differences from Scalar Form:

1. **Sign of Charges:** Unlike the scalar form where we often use only the magnitude of charges and determine direction separately, in the vector form, we **must include the signs** of q_1 and q_2. A positive product indicates repulsion, and a negative product indicates attraction, which is correctly handled by the direction of $\vec{r}$.
2. **Denominator and Numerator:** Instead of r^2 in the denominator and a unit vector $\hat{r}$ in the numerator (i.e., $k\frac{q_1 q_2}{r^2}\hat{r}$), we use r^3 in the denominator and the full position vector $\vec{r}$ in the numerator. This is because $\hat{r} = \frac{\vec{r}}{r}$, so substituting this into the standard vector form gives $k\frac{q_1 q_2}{r^2}\frac{\vec{r}}{r} = k\frac{q_1 q_2}{r^3}\vec{r}$.

Substituting Values

Now, substitute the calculated values into the vector form of Coulomb's Law:

- $k = 9 \times 10^9$ Nm^2/C^2
- $q_1 = -5$ C
- $q_2 = +5$ C
- $\vec{r} = 3\hat{i} + 4\hat{k}$
- $r = 5$

$\vec{F} = (9 \times 10^9)\frac{(-5)(+5)}{(5)^3}(3\hat{i} + 4\hat{k})$ $\vec{F} = (9 \times 10^9)\frac{-25}{125}(3\hat{i} + 4\hat{k})$ $\vec{F} = (9 \times 10^9)\left(-\frac{1}{5}\right)(3\hat{i} + 4\hat{k})$ $\vec{F} = -\frac{9\times10^9}{5}(3\hat{i} + 4\hat{k})$ N

We can also write it as: $\vec{F} = (9 \times 10^9) \left(-\frac{3}{5}\hat{i} - \frac{4}{5}\hat{k}\right)$ N

Or, by performing the division: $\vec{F} = -1.8 \times 10^9(3\hat{i} + 4\hat{k})$ N $\vec{F} = (-5.4\hat{i} - 7.2\hat{k}) \times 10^9$ N

Calculating Electric Force on a Charge

Problem Statement

Calculate the electric force on a −2 C charge due to a +4 C charge.

Solution Steps

1. **Determine the position vector $\vec{R}$:** The position vector $\vec{R}$ is drawn from the charge applying the force (source charge) to the charge on which the force is applied (test charge). Let the source charge be Q_1 = +4 C and the test charge be Q_2 = −2 C. The vector $\vec{R}$ is given by $\vec{P} - \vec{Q}$, where $\vec{P}$ is the position of the test charge and $\vec{Q}$ is the position of the source charge. Given the coordinates, $\vec{R} = (1 - 5)\hat{i} + (3 - 0)\hat{j} = -4\hat{i} + 3\hat{j}$.
2. **Calculate the magnitude of $\vec{R}$:** The magnitude of $\vec{R}$ is $R = \sqrt{(-4)^2 + (3)^2} = \sqrt{16 + 9} = \sqrt{25} = 5$.
3. **Apply Coulomb's Law in vector form:** The electric force $\vec{F}$ between two point charges Q_1 and Q_2 is given by: $\vec{F} = k\frac{Q_1 Q_2}{R^3}\vec{R}$ Where:
   - $k = 9 \times 10^9$ N m^2/C^2 (Coulomb's constant)
   - Q_1 = +4 C
   - Q_2 = −2 C
   - $\vec{R} = -4\hat{i} + 3\hat{j}$
   - $R = 5$
4. **Substitute the values into the formula:** $\vec{F} = (9 \times 10^9)\frac{(+4\text{ C})(-2\text{ C})}{(5)^3}(-4\hat{i} + 3\hat{j})$ $\vec{F} = (9 \times 10^9)\frac{-8}{125}(-4\hat{i} + 3\hat{j})$
5. **Simplify the expression:** The options provided indicate that the numerical coefficient outside the vector part is positive. This implies that the negative sign from the product $Q_1 Q_2$ should be incorporated into the vector $\vec{R}$. $\vec{F} = (9 \times 10^9)\frac{8}{125}[-(-4\hat{i} + 3\hat{j})]$ $\vec{F} = (9 \times 10^9)\frac{8}{125}(4\hat{i} - 3\hat{j})$
6. **Final Answer:** The vector part of the force is $(4\hat{i} - 3\hat{j})$. Comparing this with the options, option A is the correct answer, which has the vector component $(4\hat{i} - 3\hat{j})$. The exact numerical value of the magnitude is not required for selection based on the options provided.

Notes



Next Class Topic

In the upcoming class, the concept of the electric field will be introduced and discussed.

Homework Assignment

- **Assignment Source:** The second assignment will be posted on the Telegram channel and uploaded to the platform.
- **Questions:** Students are required to solve questions 1 to 15 from the second assignment.
- **First Assignment:** The first assignment is already available on the Telegram channel and in the previous class session notes.

Leaderboard

Score & Rank Logic

Player will get scores according to their correctness of answer plus bonus points for consistently giving correct answers. They will be ranked according to their speed.

- n 3 - 330
- n 3 - 330
- Virgil 3 - 330
- Aditya M H 3 - 230
- Shripathi 3 - 330
- Muthu Jaghan 2 - 200
- Miruthula An 2 - 200
- Astrophile 2 - 200
- B. Dhivyadarshan 2 - 200
- Saisricharan Devaraju 2 - 200
- Hemachandran 1 - 100

Notes

Did you find these notes helpful?